Freight Rate Prediction Challenge

Machine Learning Engineer Assessment | Spotter

1. Executive summary

A CatBoost regression model was trained on the 48,000 labelled development rows. A chronological holdout was used to approximate prediction on later dates. The selected iteration count was then used to refit the model on all labelled rows and generate 12,000 validation predictions plus 31 December scenario predictions.

2. Data exploration and quality

The development data contains 48,000 rows and 14 columns, covering 2025-01-01 through 2025-10-31. The target is posted_rate. The observed target mean was $2,373.98 (standard deviation $1,486.49; minimum $57.22; maximum $25,533.00). Weight had 300 missing values and 292 non-positive values. market_index had 374 missing values. Missing numeric values were left as missing for CatBoost; non-positive weights were converted to missing. load_id was excluded from model features.

3. Features and model

Categorical features: pickup, delivery, equipment. Numeric features: pickup/delivery latitude and longitude, distance, weight, market_index, quote_signal. Date was parsed and expanded into year, month, day, day of week, day of year and ISO week. CatBoostRegressor was used because it supports categorical inputs and missing numeric values without requiring one-hot encoding or imputation.

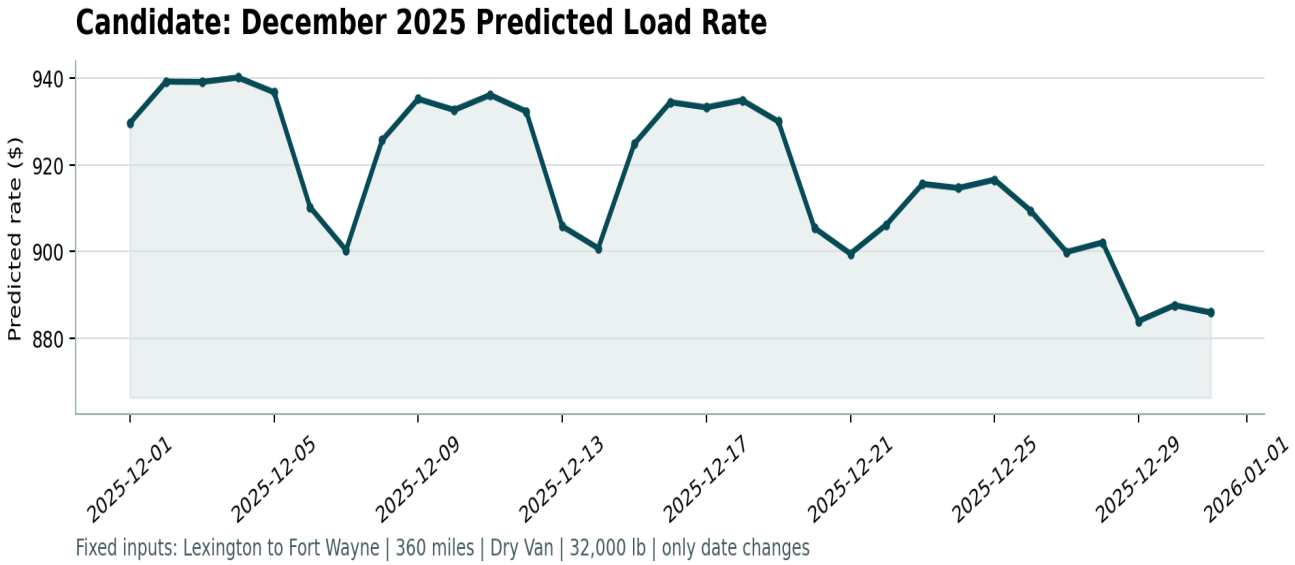
4. Validation and split

A chronological split was used. The first 80% of unique dates (training through 2025-08-31) formed the training partition (38,477 rows). The remaining dates from 2025-09-01 onward formed the holdout (9,523 rows). Early stopping selected iteration 188. Holdout metrics: MAE $113.60, RMSE $632.87, R-squared 0.8280. RMSE is notably larger than MAE, indicating a tail of larger errors; further residual and segment analysis would be useful before production deployment.

5. Final predictions

The model was refit on all 48,000 labelled records using the selected iteration count. Predictions were aligned to the supplied load_id template and checked for 12,000 unique IDs and positive finite values. The December scenario contains only pickup, delivery, distance, equipment, weight and date. Features not supplied in that scenario (coordinates, market_index and quote_signal) are treated as missing by the model. The scorer validated both outputs and generated the chart below.

6. December 2025 scenario chart



Reproducibility: see README.md and train_predict.py. Run score.py after installing requirements.txt. The provided scoring script checks output schema and scenario inputs and creates the chart; it does not calculate final hidden-set accuracy.